

Peer-Effects in Electric School Bus Adoption

Brown Bag: Applied Microeconomics (Spring 2025)

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April 11, 2026

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Why Electric School Buses (ESBs) provide a unique setting?

- **High Cost:** \$300k ESB vs \$100k diesel
- **High Uncertainty:** Range anxiety, charging logistics, cold weather performance
- **Policy Scale:** \$5 Billion announced federal funding (EPA Clean School Bus Program)
- Interesting setting for **peer-effects** (and potentially **social learning**) in the public sector

The Core Question: Contagion or Coincidence?

- We observe strong spatial clustering of adoptions
- **Is it Contagion?** Does a neighbor's choice *cause* my choice (Peer Effects)?
- **Or is it Coincidence?** Do neighbors simply share the same climate, politics, and state policies (Correlated Shocks)?

→ **See Determinants of Own Adoption**

Early Trends: Participation Map

Social Learning/Applied Micro BB (Fall 2025)/esb_adoption_by.

Two Potential Channels (The Mechanisms)

1. Emulation (Reputation)

- "Keeping up with the Joneses"
- Mechanism: **Visibility** & Rivalry
- Likely **non-negative**

2. Information from Non-Local Peers (Regional Ring)

- Reducing technical uncertainty
- Mechanism: **Similar operating conditions** beyond immediate neighbors
- Can be **positive** (success) or **negative** (discouragement)

The Identification Challenge

The Manski (1993) Reflection Problem

$$Y_i = \underbrace{\beta(WY)_i}_{\text{Endogenous}} + \underbrace{\gamma X_i + \delta(WX)_i}_{\text{Contextual}} + \underbrace{\alpha_i + \epsilon_i}_{\text{Correlated}}$$

- We see A and B adopt. Did A copy B, or did a state subsidy hit both?
- **Solution:** We need an exogenous shock that moves Peers (Y_j) but not Focal District (Y_i)

Our Solution: The EPA Lottery

- **The Program:** EPA Clean School Bus Program (Rounds 1 & 3)
- **The Mechanism:** Oversubscribed funding allocated via **LOTTERY**
- **The Data Win:** We obtained the list of **Waitlisted Applicants**
- **Randomization:** As-good-as random **within priority strata**; we condition on `is_priority`.

The Instrument (Z): For the applicant pool, a peer's lottery win is **randomly assigned**.

$$Z_j = \mathbb{I}(\text{Peer } j \text{ won lottery})$$

Validating the Design: The "Strata" Fix

Problem: Naive Balance Masks Bias

- Aggregated data looks balanced on income ($p > 0.50$).
- **BUT:** Stratified tests reveal selection bias in the "Non-Priority" pool.

Solution: Stratified Balance

- Within "Priority" buckets, the lottery is cleaner.
- Control for `is_priority` and demographics.

Identification relies on: $Z \perp \epsilon \mid$ Priority Status and other observables

→ See Balance Tables

Separating Channels: Spatial Matrices

To disentangle mechanisms, we estimate the following 2SLS model:

$$\text{Adopt}_i = \beta_E (\widehat{W_{geo} \text{ Adopt}}) + \beta_R (\widehat{W_{clim} \text{ Adopt}}) + \gamma X_i + \varepsilon_i$$

- **Emulation** (W_{geo}): KNN-6 nearest districts — visibility/rivalry.
- **Regional ring** (W_{clim}): KNN-20 *minus* KNN-6 (neighbors 7–20) — info from non-local, similar conditions. *Currently distance-based; climate-similarity W planned.*
- Charter Schools currently filtered out

Spatial Construction Details — **Sample Filtering Waterfall**

Main Results: Naive OLS vs. Causal 2SLS

	(1) Naive OLS	(2) 2SLS (IV)
Peer Effects		
Emulation ($\mathbf{W}_{geo} \cdot Y$)	0.115*** (0.019)	0.053 (0.039)
Regional Ring ($\mathbf{W}_{clim} \cdot Y$)	0.161*** (0.027)	0.152*** (0.045)
Controls		
Priority Status	0.106***	0.108***
Log(Income)	-0.020**	-0.021**
<i>SW Partial F (Geo)</i>	–	11,955
<i>SW Partial F (Ring)</i>	–	5,292

Interpretation: Instrumented estimates indicate a **causal equilibrium peer spillover** from non-local peers (regional ring); the local emulation coefficient attenuates once instrumented. Mechanism (learning vs other) left for follow-up tests.

Implications & Mechanism Roadmap

1. Policy Implication

- Priority status strongly predicts adoption (consistent with program rules)
- Peer spillovers appear economically meaningful

2. Threat 2: Measurement of "Outcome"

- **Problem:** To test learning, we need to know if a peer had a "Good" or "Bad" experience
- **Data 1 (Gold):** Survey data (currently small sample)
- **Data 2 (Silver):** Local news sentiment analysis (factiva/NexisUni)
- **Data 3 (Bronze):** Revealed Preference (Does the peer apply again in Round 3?)

Next Step: Interact $\widehat{PeerAdopt}$ with peer's *Outcome Quality* to identify discouragement.

Feedback & Next Steps

Feedback Requested

- **Peer network/definition:** Keep KNN-6 (local) + “regional ring”; add a climate-similarity W ; consider dealer/utility and border-pair links.
- **Exclusion threats:** Could neighbor wins affect me via dealer/utility capacity or state targeting (non-info channels)?
- **Discouragement:** Good pointers to negative social learning papers?

Next Steps

- **Lags/event-time:** Neighbor win/delivery at $t-1 \rightarrow$ own apply/adopt at t .
- **Fixed effects:** Year FE + state $\times$ year; (later) utility/dealer-region $\times$ year.
- **Network build:** Implement climate- W ; keep leave-one-out weights; run border-donut robustness.
- **Robustness:** Logit/probit with control-function for binary Y ; compare marginal effects to 2SLS-LPM.

Backup: Determinants of Own Adoption (OLS)

	Model 1 (Naive)	Model 2 (With Priority)	Model 3 (Interaction)
Log(Median Income)	-0.022**	-0.011	-0.012
Poverty Rate	0.177***	0.138***	0.132
Log(Enrollment)	0.026***	0.025***	0.025***
% Hispanic	-0.019	-0.024	-0.024
Priority Status	—	0.110***	0.101***
Priority $\times$ Poverty	—	—	0.061 (ns)

Key: Income effect disappears when controlling for Priority Status. EPA rules drive the income correlation.

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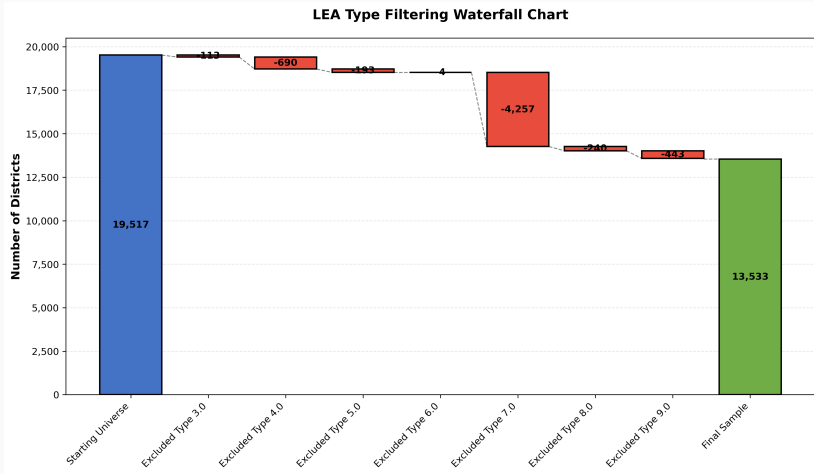
Backup: Balance Table (The "Strata" Proof)

Variable	Mean (Winners)	Mean (Losers)	Diff	P-Val
Panel A: Naive (All Applicants)				
Median Income	65,164	65,863	-698	0.534
Poverty Rate	0.14	0.14	0.00	0.511
% Hispanic	0.11	0.13	-0.02	0.004***
Panel B: Priority Applicants Only (N=1,220)				
Median Income	54,050	55,002	-952	0.425
Poverty Rate	0.17	0.16	0.00	0.432
Enrollment	10,076	6,193	3,883	0.049**
Panel C: Non-Priority Applicants Only (N=1,332)				
Median Income	68,050	81,142	-13,091	0.000***
Poverty Rate	0.13	0.10	0.03	0.000***

Takeaway: Naive balance masks heterogeneity. Priority/Non-Priority groups are distinct.

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Backup: Sample Filtering (LEA Types)



Notes: Starting universe = 19,517 districts; final analysis sample = 13,533 after excluding LEA types 3, 4, 5, 6, 7, 8, 9.

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Backup: Spatial Matrix Construction

1. Emulation Matrix ($\mathbf{W}_{geo}$)

- Calculated using K-Nearest Neighbors with $k = 6$.
- Captures the closest 6 districts (centroids).

2. Regional Ring Matrix ($\mathbf{W}_{clim}$)

- Calculated using K-Nearest Neighbors with $k = 20$.
- **Orthogonalization:** We subtract the set of neighbors in $\mathbf{W}_{geo}$ from this set.
- *Result:* A "donut" of neighbors 7 through 20.
- Ensures that β_E and β_R are identified from **distinct** sets of peers.
- *Planned robustness:* Replace with a **climate-similarity** matrix (HDD/CDD/winter lows) to test mechanism.

Backup: Peer Effects Estimates (Full)

	(1) Naive OLS	(2) 2SLS (A)	(3) 2SLS (B)
Peer Effects			
Emulation ($\mathbf{W}_{geo}$)	0.115***	0.053	0.041**
Regional Ring ($\mathbf{W}_{clim}$)	0.161***	0.152***	-0.025
Select Controls			
Priority Status	0.106***	0.108***	0.052***
Own Lottery Win (IV_Z)	–	–	0.962***
First Stage F (Geo)	–	11,955	11,564
First Stage F (Ring)	–	5,292	5,173
R-Squared	0.072	0.071	0.779

Discussion: Model (3) includes Own Lottery Win in the outcome—*not preferred* for peer-effects identification; shown only as a sensitivity. Main specification is Model (2) with instruments $\{\mathbf{W}_{geo}Z, \mathbf{W}_{clim}Z\}$ and no own- Z control.

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